Create a comparison table listing at least 3 key differences between WAN and LAN, and give one real-world example of each from apps or services you use daily (for example, IRCTC, Instagram, or Zomato).
ANS:

Comparison Table: LAN vs WAN

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
1. Coverage Area	Covers a small area, such as a home, office, or college.	Covers a large area, such as cities, countries, or the world.
2. Speed	Usually offers high speed within the local network.	Speed depends on the internet connection and network infrastructure.
3. Ownership	Usually managed by an individual or organization.	Often uses infrastructure managed by multiple service providers.
4. Cost	Generally cheaper to set up and maintain.	Generally more expensive to build and maintain.
5. Example	Office computers and printers connected to the same Wi-Fi router.	Instagram or Zomato accessed over the internet.

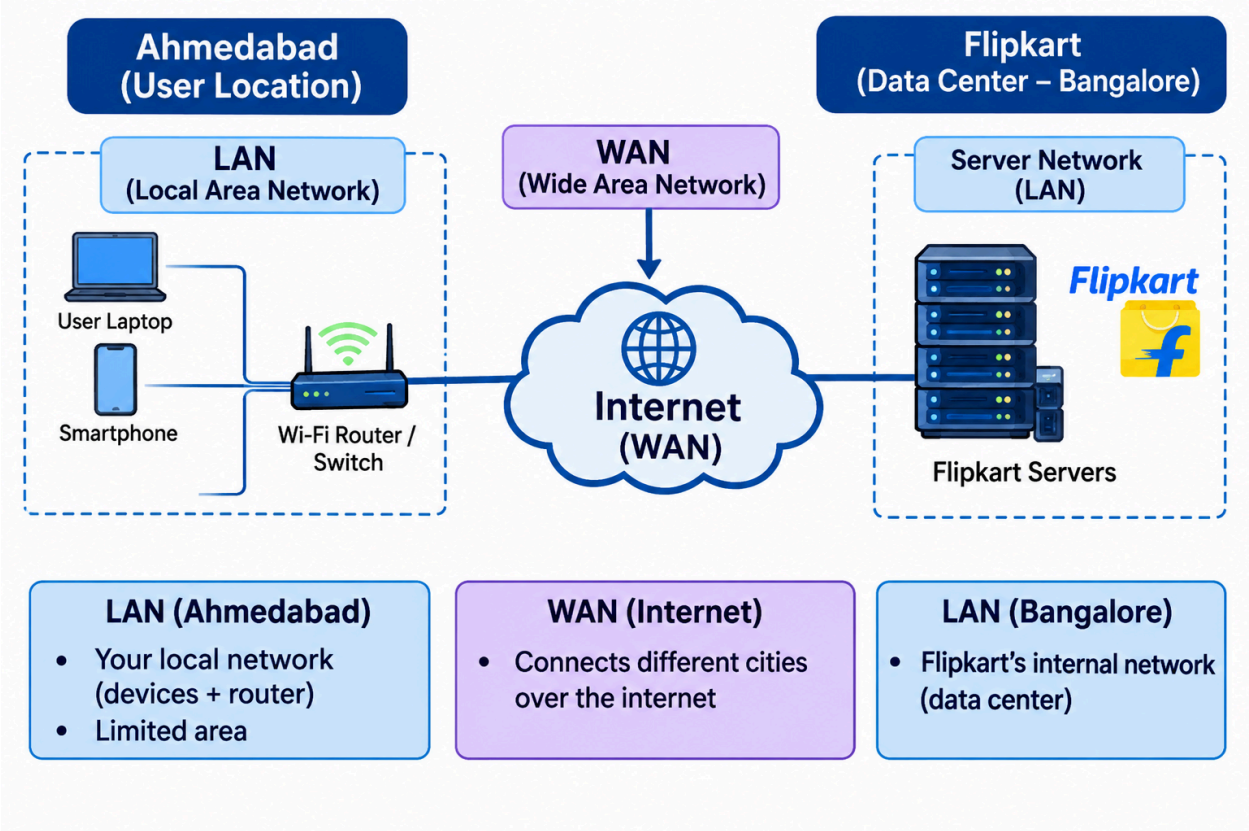
Real-world examples

- LAN: In an office, computers and a printer connect to the same local network to share files and print documents.
- WAN: When you use Instagram to upload photos or Zomato to order food, your device communicates with remote servers over the internet, which is a WAN.

Conclusion: LAN connects devices within a limited area, while WAN connects networks over large geographical distances.

Draw a simple network diagram using any drawing tool (or on paper and upload a photo) that shows how a user in Ahmedabad could connect to a Flipkart data center in Bangalore using a WAN, labeling the LAN and WAN segments clearly.

ANS:



Here is the network diagram showing how a user in Ahmedabad connects to Flipkart's data center in Bangalore through a WAN.

LAN in Ahmedabad: Laptop and smartphone connect to a Wi-Fi router.

WAN: The internet connects the Ahmedabad network to the remote data center.

LAN in Bangalore: Flipkart's servers communicate over their internal network.

Conclusion: A LAN connects devices within a limited area, whereas a WAN connects networks across large geographical distances. When the user opens Flipkart, the request travels through the internet to its servers and the response returns to the user.

Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

ANS:

MPLS (Multiprotocol Label Switching) is a networking technology that forwards data packets using short labels instead of relying on IP address lookups at every router.

1. It assigns a label to packets when they enter the MPLS network.
2. Routers forward packets based on these labels along a predefined path.
3. Traditional IP routing uses destination IP addresses to make forwarding decisions at each router.
4. MPLS can improve forwarding efficiency and provide more predictable traffic paths.
5. It supports Quality of Service (QoS), helping prioritize important traffic such as voice and video calls.
6. MPLS is widely used by enterprises and service providers to connect branch offices securely and reliably.

Sources: Cisco — What is MPLS? and Cisco — Introduction to MPLS.

Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing.

Hint: Look for how labels are used and how packets move between routers.

ANS:

1. Label-Based Forwarding: In MPLS, routers forward packets using short labels instead of checking the destination IP address at every hop, unlike traditional IP routing.

2. Predefined Path: MPLS packets follow a Label-Switched Path (LSP) established through the network, whereas standard IP routing selects the next hop using the routing table at each router.

List three specific benefits MPLS offers over traditional routing for companies running high-traffic apps like YouTube or BookMyShow, and briefly explain how each benefit helps in real-world scenarios.

ANS:

Three Benefits of MPLS over Traditional IP Routing

1. Better Traffic Management: MPLS allows companies to prioritize important traffic. For example, BookMyShow can prioritize payment transactions to help maintain a smooth booking experience during high-demand ticket sales.

2. Improved Quality of Service (QoS): MPLS supports prioritization of delay-sensitive traffic. For example, YouTube can benefit from prioritizing video traffic on managed network links to reduce buffering and delays.

3. Reliable and Predictable Routing: MPLS can direct traffic along predefined paths. For example, a company with multiple offices can use MPLS to maintain consistent connectivity between branches and central servers during busy periods.

Conclusion: MPLS helps businesses manage network traffic efficiently, prioritize critical applications, and deliver more predictable network performance. However, actual performance depends on network design, capacity, and service-provider configuration.

In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.
ANS:

Cisco Packet Tracer: Two Branch Office Networks

Objective: Create two separate branch office LANs using two routers and configure IP addresses so devices in each LAN can ping one another.

1. Network topology

Branch Office 1 — LAN 1

PC1 (`192.168.1.2`) · PC2 (`192.168.1.3`)

Switch 1

Router 1 — R1 (`192.168.1.1`)

Optional WAN link between routers

Router 2 — R2 (`192.168.2.1`)

Switch 2

Branch Office 2 — LAN 2

PC3 (`192.168.2.2`) · PC4 (`192.168.2.3`)

2. IP addressing table

Device	IP address	Subnet mask
R1 LAN interface	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
R2 LAN interface	192.168.2.1	255.255.255.0
PC3	192.168.2.2	255.255.255.0
PC4	192.168.2.3	255.255.255.0

3. Configuration steps

1. Open Cisco Packet Tracer.
2. Add two routers, two switches, and four PCs.
3. Connect each PC to its branch switch, and connect each switch to its router using copper straight-through cables.
4. Configure each PC by opening Desktop → IP Configuration and entering its IP address and subnet mask.
5. Configure router interfaces through the CLI. Interface names may vary by router model.

Router 1 (R1):

```
None
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
end
```

Router 2 (R2):

```
None
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
end
```

4. Test connectivity

On PC1, open Desktop → Command Prompt and run:

```
None
ping 192.168.1.3
```

On PC3, run:

```
None
ping 192.168.2.3
```

Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.

Hint: Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.
ANS:

Configure a GRE Site-to-Site Tunnel in Cisco Packet Tracer

Objective: Connect two branch offices using GRE tunnel interfaces so that PCs in one LAN can ping PCs in the other LAN.

Note: GRE creates a tunnel but does not encrypt traffic by itself. This is a GRE site-to-site tunnel, not a fully encrypted IPsec VPN.

1. Network topology

Branch 1 LAN

PC1: `192.168.1.2/24`

Gateway: `192.168.1.1`

Router R1

LAN: `192.168.1.1/24`

WAN: `10.0.0.1/30`

Tunnel0: `172.16.0.1/30`

GRE Tunnel over WAN

`172.16.0.1 ↔ 172.16.0.2`

Router R2

WAN: `10.0.0.2/30`

LAN: `192.168.2.1/24`

Tunnel0: `172.16.0.2/30`

Branch 2 LAN

PC2: `192.168.2.2/24`

Gateway: `192.168.2.1`

2. Configure router interfaces

Use a separate physical WAN link between R1 and R2. The interface names below are examples; adapt them to your router model and cabling.

Router R1

```
None
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

Router R2

```
None
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
```

```
exit
```

3. Create the GRE tunnel

On R1:

```
None
interface tunnel 0
ip address 172.16.0.1 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 10.0.0.2
no shutdown
exit
```

On R2:

```
None
interface tunnel 0
ip address 172.16.0.2 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 10.0.0.1
no shutdown
exit
```

4. Add routes between branch LANs

On R1:

```
None
ip route 192.168.2.0 255.255.255.0 172.16.0.2
```

On R2:

```
None
ip route 192.168.1.0 255.255.255.0 172.16.0.1
```

5. Configure the PCs

Setting	PC1	PC2
IP address	192.168.1.2	192.168.2.2
Subnet mask	255.255.255.0	255.255.255.0
Default gateway	192.168.1.1	192.168.2.1

6. Verify the tunnel and connectivity

On both routers, run:

```
None
show ip interface brief
show interfaces tunnel 0
show ip route
```

From PC1, test:

```
None
ping 192.168.2.2
```

From PC2, test:

```
None
ping 192.168.1.2
```

Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.

Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.
ANS:

Remote Access VPN Simulation in Cisco Packet Tracer

Objective: Configure a VPN client PC to connect securely to a branch office router using a username and password.

1. Network topology

Remote User PC

IP: `192.168.10.2`

VPN Client enabled

Internet / WAN

VPN connection

Branch Router

VPN Server IP: `10.0.0.1`

Branch Office LAN

Server: `192.168.1.10`

2. Configure the branch router

In a compatible Cisco IOS environment, configure a basic IPsec remote-access VPN using IKEv1 and a pre-shared key.

Example settings:

- VPN server address: `10.0.0.1`
- VPN username: `remoteuser`
- VPN password: `Remote@123`
- Pre-shared key: `VPNkey123`

Important: These are example credentials for a lab only. The actual router configuration depends on the IOS image and Packet Tracer version.

3. Configure the VPN client PC

1. Open the remote PC in Packet Tracer.
2. Go to Desktop → VPN if the VPN tab is available.
3. Enter the VPN server IP address: `10.0.0.1`.
4. Enter the username: `remoteuser`.
5. Enter the password: `Remote@123`.
6. Enter the pre-shared key if the client requests it.
7. Click Connect.

4. Verify the connection

After connecting, check whether the VPN client receives the expected VPN address. Then open the PC's Command Prompt and test access to the branch server:

```
None
ping 192.168.1.10
```

Expected result: Successful replies indicate that the remote PC can reach the branch office server through the VPN connection.

5. Conclusion

A remote-access VPN allows employees to connect to an organization's private network from a remote location. Authentication and encryption help protect data while it travels over an untrusted network.

Packet Tracer limitation: The built-in PC VPN tab and remote-access VPN server support depend on the version and device model. If the VPN tab is unavailable, or the router does not support the required IPsec commands, this exact scenario may need to be implemented in a compatible IOS lab or GNS3 instead.

List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.
ANS:

Difference Between Site-to-Site VPN and Remote Access VPN

Feature	Site-to-Site VPN	Remote Access VPN
Purpose	Connects two branch office networks.	Connects an individual remote user to an office network.
Configuration	Configured on both branch routers using GRE tunnels and routing.	Configured on a VPN client PC and a compatible VPN server/router.
Users	Connects multiple devices across both LANs.	Connects individual users or devices.
Example	Ahmedabad branch connected to Bangalore branch.	An employee accessing office resources from home.
Authentication	Often uses IPsec keys or certificates, depending on configuration.	Commonly uses user credentials and encryption, depending on configuration.

Advantages and disadvantages

1. Site-to-Site VPN

Advantages

- Connects entire branch networks, allowing multiple devices to communicate.
- Provides continuous connectivity between offices without requiring each employee to start a separate VPN session.

Disadvantages

- Requires configuration and maintenance at both ends.
- Troubleshooting can be complex when routing, tunnel interfaces, or WAN links fail.

2. Remote Access VPN

Advantages

- Allows employees to access office resources from home or while travelling.
- Individual user authentication helps control who can access the corporate network.

Disadvantages

- Each user needs a compatible VPN client and appropriate configuration.
- Many simultaneous connections can increase server load and require additional management.

Conclusion

A site-to-site VPN is suitable for connecting office networks, while a remote access VPN is suitable for connecting individual employees to an office network. In Packet Tracer, the GRE tunnel demonstrates network-to-network connectivity; however, GRE alone does not encrypt traffic. A secure VPN requires suitable encryption, such as IPsec. Remote-access VPN features depend on the Packet Tracer version and supported devices.

Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.
ANS:

Site-to-Site GRE VPN Configuration Report

Software: Cisco Packet Tracer
Objective: Connect two branch office LANs using GRE tunnel interfaces and static routing so PCs at both branches can communicate.

1. Network IP addressing

Device / Interface	IP address	Subnet mask
R1 LAN interface	192.168.1.1	255.255.255.0

R1 WAN interface	10.0.0.1	255.255.255.252
R1 Tunnel0	172.16.0.1	255.255.255.252
R2 WAN interface	10.0.0.2	255.255.255.252
R2 LAN interface	192.168.2.1	255.255.255.0
R2 Tunnel0	172.16.0.2	255.255.255.252
PC1	192.168.1.2	255.255.255.0
PC2	192.168.2.2	255.255.255.0

PC1 default gateway: `192.168.1.1`

PC2 default gateway: `192.168.2.1`

Note: The interface names below assume GigabitEthernet 0/0 is the LAN interface and GigabitEthernet 0/1 is the WAN interface.

2. Router R1 configuration

Enter the following commands in the R1 CLI.

```
None

enable
configure terminal

! Configure LAN interface
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

! Configure WAN interface
interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit

! Configure GRE tunnel
interface tunnel 0
ip address 172.16.0.1 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 10.0.0.2
no shutdown
exit

! Route to Branch 2 LAN
ip route 192.168.2.0 255.255.255.0 172.16.0.2

end
write memory
```

3. Router R2 configuration

Enter the following commands in the R2 CLI.

```
None
```

```
enable
configure terminal

! Configure LAN interface
interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

! Configure WAN interface
interface gigabitEthernet 0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

! Configure GRE tunnel
interface tunnel 0
ip address 172.16.0.2 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 10.0.0.1
no shutdown
exit

! Route to Branch 1 LAN
ip route 192.168.1.0 255.255.255.0 172.16.0.1

end
write memory
```

4. Verification commands

Run these commands on both routers to verify interface and routing status.

None

```
show ip interface brief
```

```
show interfaces tunnel 0
show ip route
```

Test the tunnel endpoints from R1:

```
None

ping 172.16.0.2
```

Test end-to-end connectivity from PC1:

```
None

ping 192.168.2.2
```

Test connectivity in the reverse direction from PC2:

```
None

ping 192.168.1.2
```

Expected result: Successful ping replies confirm that the PCs at both branch offices can communicate through the GRE tunnel.

5. Conclusion

The site-to-site GRE tunnel was configured using physical LAN/WAN interfaces, `interface tunnel 0`, tunnel source and destination commands, and static routes. These settings provide connectivity between the two branch networks.

Security note: GRE encapsulates packets but does not encrypt them. To create an encrypted site-to-site VPN, IPsec must also be configured and supported by the selected router model and Packet Tracer version.

In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.

ANS:

Cisco Packet Tracer: Static NAT Configuration

Objective: Create a network with one router, one internal PC, and one external PC. Configure static Network Address Translation (NAT) so the internal PC can communicate with the external PC.

1. Network topology

Internal Network (LAN)

PC1 — Private IP: `192.168.1.10`

Router R1

LAN: `192.168.1.1`

WAN: `203.0.113.1`

Static NAT: `192.168.1.10` ↔ `203.0.113.10`

External Network

PC2 — Public test IP: `203.0.113.20`

The `203.0.113.0/24` range is reserved for documentation and lab examples. It is not a real, publicly routable internet address range.

2. IP addressing table

Device	Interface	IP address	Subnet mask

PC1 (Internal)	NIC	192.168.1.10	255.255.255.0
Router R1	G0/0 (LAN)	192.168.1.1	255.255.255.0
Router R1	G0/1 (WAN)	203.0.113.1	255.255.255.0
PC2 (External)	NIC	203.0.113.20	255.255.255.0

Set PC1's default gateway to `192.168.1.1` and PC2's default gateway to `203.0.113.1`.

3. Configure router interfaces

Open the router CLI and enter:

```
None

enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet 0/1
ip address 203.0.113.1 255.255.255.0
ip nat outside
no shutdown
exit
```

4. Configure static NAT

Enter this command on the router:

```
None

ip nat inside source static 192.168.1.10 203.0.113.10
end
write memory
```

This creates a permanent mapping:

- Inside local: `192.168.1.10` — the internal PC's private address.
- Inside global: `203.0.113.10` — the public-style address representing the internal PC.

5. Test the network

On PC1, open Desktop → Command Prompt and run:

```
None
ping 203.0.113.20
```

On the router, verify the NAT configuration:

```
None
show ip nat translations
show ip nat statistics
show ip interface brief
```

Generate traffic first before checking the translation table, as some entries may appear only after traffic has been sent.

Expected result: PC1 should be able to communicate with PC2, and the router should show the static NAT mapping.

If the ping fails, check the interface status, cabling, IP addresses, default gateways, and whether the router supports NAT in your Packet Tracer version.

6. Conclusion

Static NAT maps one private IP address to one fixed translated address. It allows the internal PC to use a consistent translated identity when communicating with external networks. Static NAT does not itself provide encryption or firewall protection.

Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address.

Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.
ANS:

Cisco Packet Tracer: Configure PAT (NAT Overload)

Objective: Configure Port Address Translation (PAT) so that two internal PCs can access a simulated public web server using a single public IP address.

1. Network topology

Internal LAN

PC1: `192.168.1.10`

PC2: `192.168.1.11`

Subnet: `192.168.1.0/24`

Router R1

LAN: `192.168.1.1`

WAN: `198.51.100.1`

PAT: One public IP + different port numbers

Simulated Public Web Server

IP: `198.51.100.10`

HTTP service: Enabled

The `198.51.100.0/24` address range is reserved for documentation and lab examples, not real public internet use.

2. IP addressing table

Device	Interface	IP address	Subnet mask
PC1	NIC	192.168.1.10	255.255.255.0
PC2	NIC	192.168.1.11	255.255.255.0
Router R1	G0/0 (LAN)	192.168.1.1	255.255.255.0
Router R1	G0/1 (WAN)	198.51.100.1	255.255.255.0
Web Server	NIC	198.51.100.10	255.255.255.0

Configure the default gateways:

- PC1 and PC2: `192.168.1.1`
- Web Server: `198.51.100.1`

3. Configure the router interfaces

Open the router CLI and enter:

```
None

enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet 0/1
ip address 198.51.100.1 255.255.255.0
ip nat outside
no shutdown
exit
```

4. Configure PAT using overload

Enter the following commands on R1:

```
None

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet 0/1 overload

end
write memory
```

Explanation:

- `access-list 1` identifies the internal private network allowed to use NAT.
- `ip nat inside` marks the LAN interface.
- `ip nat outside` marks the WAN interface.
- `overload` allows multiple internal devices to share the same public IP by using different source port numbers.

5. Enable the web server

1. Click the server in Packet Tracer.
2. Open Desktop → IP Configuration and assign `198.51.100.10/24`.
3. Set its default gateway to `198.51.100.1`.
4. Open Services → HTTP and turn HTTP On.

6. Test PAT

From PC1, open Desktop → Command Prompt:

```
None
ping 198.51.100.10
```

Repeat the ping from PC2.

Then open Desktop → Web Browser on each PC and visit:

```
None
http://198.51.100.10
```

On R1, check the NAT translations:

```
None
show ip nat translations
show ip nat statistics
```

Generate traffic from both PCs before checking the table. You should see translations associated with the same outside interface IP, with different port numbers when the traffic requires them.

7. Expected result

PC1 accesses the simulated web server.

PC2 accesses the same web server.

Both PCs share one translated public IP address through PAT.

Conclusion: PAT, also called NAT overload, allows multiple devices with private IP addresses to access external services using a single public IP address. Port numbers help the router distinguish each device's connections and return traffic to the correct PC.

Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping

ANS:

NAT Configuration Testing in Cisco Packet Tracer

1. Send a ping from the internal PC

1. Click the internal PC (PC1).
2. Open Desktop → Command Prompt.
3. Enter the external PC's IP address:

```
None
ping 203.0.113.20
```

Expected result: Successful replies confirm connectivity between the internal and external PCs.

2. Capture the NAT translation table

Click the router and open CLI. Enter:

```
None
enable
show ip nat translations
show ip nat statistics
```

For the static NAT configuration, the table should show a mapping similar to this:

Inside global	Inside local	Type

203.0.113.10	192.168.1.10	Static

These addresses correspond to the example configuration from the previous task.

3. Verify the result

- Ping successful: The internal PC can reach the external PC.
- NAT mapping visible: The router translates the private address to the configured global address.
- No dynamic entry visible: This can be normal for static NAT because its mapping is permanent. For PAT, generate traffic from both PCs before checking the table.

Submission note: I cannot capture a live screenshot from your Packet Tracer session. Run the commands above and take a screenshot showing the ping replies and the router's NAT translation table for your assignment.

Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

ANS:

Cisco Packet Tracer: Add a Third PC and Configure PAT

Objective: Connect three internal PCs to an external network using one public IP address with PAT (NAT Overload).

1. Update the network topology

Internal LAN — Switch

PC1: `192.168.1.10`

PC2: `192.168.1.11`

PC3: `192.168.1.12`

Gateway: `192.168.1.1`

Router R1

LAN: `192.168.1.1`

WAN: `198.51.100.1`

PAT enabled

External Web Server

`198.51.100.10`

2. Configure the third PC

Connect PC3 to the internal switch using a copper straight-through cable.

Open PC3 → Desktop → IP Configuration and enter:

Setting	Value
IP address	`192.168.1.12`
Subnet mask	`255.255.255.0`
Default gateway	`192.168.1.1`

Keep PC1 and PC2 configured with their existing addresses.

3. Update the router's PAT configuration

On Router R1, enter:

```
None
enable
```

```
configure terminal

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet 0/1 overload

end
write memory
```

This access list already covers all three PCs because their addresses belong to `192.168.1.0/24`. If PAT was configured previously with this same rule, no NAT changes are required.

Verify that the LAN and WAN interfaces still have `ip nat inside` and `ip nat outside`, respectively.

4. Test connectivity from all three PCs

Open Desktop → Command Prompt on each PC and run:

PC1:

```
None
ping 198.51.100.10
```

PC2:

```
None
ping 198.51.100.10
```

PC3:

```
None
ping 198.51.100.10
```

5. Verify the NAT translation table

On R1, run:

```
None
show ip nat translations
show ip nat statistics
```

Generate traffic from each PC before checking the table. For PAT, you should see translations resembling the following example:

Inside local	Inside global
192.168.1.10	198.51.100.1
192.168.1.11	198.51.100.1
192.168.1.12	198.51.100.1

The actual table includes protocol and port information for active PAT sessions. All three PCs share the same translated IP, while port numbers distinguish their connections. ICMP translations may appear differently depending on router implementation.

6. Result

Expected outcome: All three internal PCs can reach the external web server, and the router's NAT table shows translations for traffic generated by each PC.

For your assignment, capture screenshots of the three successful ping tests and the output of `show ip nat translations`.

Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

ANS:

How NAT and PAT Help Apps Like Zomato and Flipkart

NAT (Network Address Translation) converts private IP addresses into public IP addresses so devices can communicate with external networks.

PAT (Port Address Translation) allows multiple devices to share a single public IP address by using different port numbers to identify their connections.

How they help in real life

1. Efficient use of public IP addresses: Apps like Zomato and Flipkart rely on large networks of servers. NAT and PAT help organizations connect many internal devices using a limited number of public IP addresses.

2. Handling multiple connections: PAT distinguishes simultaneous connections using port numbers. This helps many employees or devices in an organization access cloud services and the internet through a shared public IP.

Two real-world scenarios where incorrect NAT/PAT setup can break connectivity

Scenario 1: Customers cannot access the application

If a company's NAT configuration is incorrect, requests from its internal application servers may fail to reach external services, such as payment gateways or delivery-tracking APIs. As a result, transactions or tracking updates may fail.

Scenario 2: Website or server becomes unreachable

If a business configures incorrect port forwarding or NAT rules, incoming connections may not reach the correct web server. For example, a Flipkart-like shopping website might become inaccessible to users because HTTP/HTTPS traffic is not forwarded to the correct server.

Conclusion

NAT conserves public IP addresses, while PAT enables multiple connections to share one public IP using port numbers. Correct configuration is essential for reliable connectivity and smooth application performance.

Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.
ANS:

Cisco Packet Tracer: WAN + VPN Tunnel + NAT

Objective: Connect two branch offices through a WAN using a GRE VPN tunnel, and configure NAT/PAT on the edge routers so both branches can access a simulated internet network.

The topology uses:

- Branch 1: LAN `192.168.1.0/24`
- Branch 2: LAN `192.168.2.0/24`
- GRE tunnel: Connects the two branch routers.
- NAT/PAT: Allows each branch to share its edge router's WAN IP for internet access.
- Internet server: A simulated external web server in Packet Tracer.

1. Network topology

Branch 1 LAN

PC1: `192.168.1.10`

Gateway: `192.168.1.1`

R1 — Edge Router

LAN: `192.168.1.1`

WAN: `10.0.0.1`

WAN / GRE Tunnel

R1 `10.0.0.1` ↔ R2 `10.0.0.2`

Tunnel: `172.16.0.1` ↔ `172.16.0.2`

R2 — Edge Router

WAN: `10.0.0.2`

LAN: `192.168.2.1`

Branch 2 LAN

PC2: `192.168.2.10`

Gateway: `192.168.2.1`

Connect each edge router's additional WAN interface to an ISP router, which connects to the simulated internet server. This internet path is separate from the branch-to-branch tunnel.

2. IP addressing plan

Device	Interface / device	IP address
R1	LAN G0/0	`192.168.1.1/24`
R1	WAN G0/1	`203.0.113.2/30`
R1	Tunnel0	`172.16.0.1/30`
R2	LAN G0/0	`192.168.2.1/24`
R2	WAN G0/1	`198.51.100.2/30`
R2	Tunnel0	`172.16.0.2/30`

ISP	G0/0 (to R1)	`203.0.113.1/30`
ISP	G0/1 (to R2)	`198.51.100.1/30`
ISP	G0/2 (server LAN)	`192.0.2.1/24`
Web server	NIC	`192.0.2.10/24`

These public-looking IP ranges are reserved for documentation and lab simulations, not real public internet access.

3. Configure edge router R1

Connect R1's LAN to the Branch 1 switch, and its WAN interface to the ISP router.

```
None

enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet 0/1
ip address 203.0.113.2 255.255.255.252
ip nat outside
no shutdown
```

```
exit

interface tunnel 0
ip address 172.16.0.1 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 198.51.100.2
no shutdown
exit

ip route 192.168.2.0 255.255.255.0 172.16.0.2
ip route 0.0.0.0 0.0.0.0 203.0.113.1

access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet 0/1 overload

end
write memory
```

4. Configure edge router R2

None

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet 0/1
ip address 198.51.100.2 255.255.255.252
ip nat outside
no shutdown
exit
```

```
interface tunnel 0
ip address 172.16.0.2 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 203.0.113.2
no shutdown
exit

ip route 192.168.1.0 255.255.255.0 172.16.0.1
ip route 0.0.0.0 0.0.0.0 198.51.100.1

access-list 1 permit 192.168.2.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet 0/1 overload

end
write memory
```

5. Configure the ISP router

The ISP router provides the WAN transit between both edge routers and a separate network for the simulated web server.

None

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 203.0.113.1 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
ip address 198.51.100.1 255.255.255.252
no shutdown
exit
```

```
interface gigabitEthernet 0/2
ip address 192.0.2.1 255.255.255.0
no shutdown
exit

end
write memory
```

No extra ISP routes are needed for these directly connected networks.

6. Configure PCs and the web server

Device	IP address	Default gateway
Branch 1 PC	`192.168.1.10/24`	`192.168.1.1`
Branch 2 PC	`192.168.2.10/24`	`192.168.2.1`
Web server	`192.0.2.10/24`	`192.0.2.1`

On the server, go to Services → HTTP and turn HTTP on.

7. Verify the configuration

Run these commands on R1 and R2:

```
None
```

```
show ip interface brief
show interfaces tunnel 0
show ip route
show ip nat translations
show ip nat statistics
```

Test from Branch 1 PC:

```
None
ping 172.16.0.2
ping 192.168.2.10
ping 192.0.2.10
```

Test from Branch 2 PC:

```
None
ping 172.16.0.1
ping 192.168.1.10
ping 192.0.2.10
```

Finally, open the web browser on either PC and visit `http://192.0.2.10`.

Expected results:

- The GRE tunnel carries traffic between the branch LANs.
- Both PCs can access the simulated web server through their edge routers' PAT configurations.
- NAT translations appear after the PCs generate outbound traffic.

Important: GRE provides tunneling, not encryption. For a genuinely secure site-to-site VPN, configure IPsec as well. Also, some Packet Tracer router models do not support GRE or NAT commands; use a compatible model if a command is rejected.

Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address.

Hint: Use the 'overload' keyword in your NAT configuration.

ANS:

Cisco Packet Tracer: Configure PAT for Three Devices

Objective: Allow three internal devices to access a simulated internet server using a single public IP address with PAT (NAT Overload).

1. Network topology

Internal LAN

PC1: `192.168.1.10`

PC2: `192.168.1.11`

PC3: `192.168.1.12`

Gateway: `192.168.1.1`

Edge Router R1

LAN: `192.168.1.1`

WAN: `203.0.113.2`

PAT: `overload`

ISP Router → Internet Server

Server: `192.0.2.10`

The addresses `203.0.113.0/24` and `192.0.2.0/24` are reserved for documentation and lab examples.

2. Configure the router interfaces

On R1, assuming `G0/0` is the LAN interface and `G0/1` is the WAN interface:

```
None

enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet 0/1
ip address 203.0.113.2 255.255.255.252
ip nat outside
no shutdown
exit
```

3. Configure PAT using overload

Enter these commands on R1:

```
None

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet 0/1 overload

ip route 0.0.0.0 0.0.0.0 203.0.113.1

end
write memory
```

The ISP router must have WAN address `203.0.113.1/30` on its interface connected to R1.

Explanation:

- `access-list 1` identifies all devices in the internal subnet.
- `ip nat inside` marks the LAN interface.
- `ip nat outside` marks the WAN interface.
- `overload` allows multiple devices to share the WAN interface's IP address using different port numbers.
- The default route sends internet-bound traffic to the ISP router.

4. Configure the three PCs

Set these addresses in Desktop → IP Configuration.

Device	IP address	Subnet mask	Default gateway
PC1	`192.168.1.10`	`255.255.255.0`	`192.168.1.1`
PC2	`192.168.1.11`	`255.255.255.0`	`192.168.1.1`
PC3	`192.168.1.12`	`255.255.255.0`	`192.168.1.1`

Configure the simulated web server as `192.0.2.10/24`, with default gateway `192.0.2.1`. Enable HTTP under Services → HTTP. Ensure the ISP router has a connected interface in `192.0.2.0/24`.

5. Test connectivity

From each PC, open Desktop → Command Prompt and run:

```
None
ping 192.0.2.10
```

Then open the PC's web browser and visit:

```
None
http://192.0.2.10
```

On R1, check the translations:

```
None

show ip nat translations
show ip nat statistics
```

Generate traffic from all three PCs before checking the table. Their active translations should share the public address `203.0.113.2`; port numbers or protocol identifiers distinguish individual sessions.

6. Expected result

- All three PCs can access the simulated web server.
- PAT translates their private addresses to one shared WAN IP address.
- The router's NAT table displays translations for the generated traffic.

Conclusion: PAT enables multiple internal devices to access external networks using a single public IP address, reducing public IP address requirements while tracking separate connections.

Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.
ANS:

Cisco Packet Tracer: Troubleshoot a VPN Pre-Shared Key Mismatch

Objective: Intentionally configure different pre-shared keys on two routers, identify the VPN failure, correct the keys, and verify connectivity.

Note: This exercise applies to an IPsec VPN using IKEv1. A GRE tunnel alone does not use a pre-shared key. Some Packet Tracer router models may not support all IPsec commands.

1. Intentionally misconfigure the keys

On Router R1, configure a key for Router R2:

```
None
enable
configure terminal
crypto isakmp key VPN123 address 198.51.100.2
end
```

On Router R2, intentionally configure a different key:

```
None
enable
configure terminal
crypto isakmp key WRONG456 address 203.0.113.2
end
```

The keys do not match, so IKE authentication should fail when the routers attempt to establish the IPsec VPN.

2. Identify the problem

On both routers, run:

```
None
show crypto isakmp sa
show running-config
```

Look for:

- ISAKMP SA status: The VPN may remain in a negotiation state instead of reaching `QM_IDLE`, which indicates an established IKEv1 Phase 1/Phase 2 negotiation.
- Running configuration: Check the `crypto isakmp key` entries and confirm that each key is associated with the correct peer IP address.

For security, some IOS versions conceal pre-shared key values in command output. You may need to re-enter the intended key rather than compare displayed values.

3. Fix the key mismatch

Configure the same key on R1:

```
None
enable
configure terminal
no crypto isakmp key VPN123 address 198.51.100.2
crypto isakmp key VPN123 address 198.51.100.2
end
```

Configure the matching key on R2:

```
None
enable
configure terminal
no crypto isakmp key WRONG456 address 203.0.113.2
crypto isakmp key VPN123 address 203.0.113.2
end
```

The key must match on both routers, and each peer address must be correct.

4. Verify the VPN

Generate traffic between the protected LANs to trigger negotiation, then run:

```
None
show crypto isakmp sa
show crypto ipsec sa
show crypto isakmp policy
```

If supported, the IKEv1 security association should reach `QM_IDLE`. The IPsec counters should increase when protected traffic passes through the tunnel.

Finally, test connectivity between the branch PCs:

```
None
ping 192.168.2.10
```

5. Result

Before correction: The mismatched pre-shared keys prevent successful IKE authentication, so the IPsec VPN cannot establish correctly.

After correction: Matching keys allow authentication to proceed. If the remaining IKE/IPsec policies, peer addresses, routing, and traffic selectors are correct, the VPN should establish and branch-to-branch connectivity should work.

Conclusion: `show crypto isakmp sa` helps diagnose IKE negotiation status, while `show running-config` helps inspect VPN configuration. A pre-shared key mismatch is one possible cause of VPN failure, but it is not the only one.

Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

ANS:

Troubleshooting: Branch-to-Branch VPN Connectivity

Problem: Branch 1 can ping the internet, but it cannot access devices in Branch 2 through the VPN tunnel.

1. Check connectivity and VPN status

Run these commands on both routers:

```
None
show ip interface brief
show ip route
show interfaces tunnel 0
show crypto isakmp sa
show crypto ipsec sa
show running-config
```

- `show ip route` — check routes to the other branch's LAN.
- `show interfaces tunnel 0` — verify the tunnel is up.
- `show crypto isakmp sa` — check IKE security association status.
- `show crypto ipsec sa` — check whether IPsec is encrypting and decrypting traffic.
- `show running-config` — inspect tunnel, routing, and VPN settings.

2. Identify the problem

Suppose Branch 1 uses `192.168.1.0/24` and Branch 2 uses `192.168.2.0/24`.

The issue is a missing route to Branch 2's LAN on Router 1. Internet access works because Router 1 already has a default route.

3. Fix the configuration

On Router 1, add a route through the VPN tunnel:

```
None
enable
configure terminal
ip route 192.168.2.0 255.255.255.0 172.16.0.2
end
write memory
```

On Router 2, ensure the reverse route exists:

```
None
enable
configure terminal
ip route 192.168.1.0 255.255.255.0 172.16.0.1
end
write memory
```

Assumption: The GRE tunnel addresses are `172.16.0.1` on Router 1 and `172.16.0.2` on Router 2.

4. Verify the fix

From a Branch 1 PC, ping a Branch 2 PC, for example:

```
None
ping 192.168.2.10
```

Then verify the routing table:

```
None
show ip route
show interfaces tunnel 0
```

Expected result: The ping succeeds, and the routing table contains a route to the remote branch through the tunnel.

Conclusion: The problem was a missing route to the remote branch LAN. Adding the static route restored branch-to-branch connectivity while preserving internet access. If the tunnel is down or IPsec counters do not increase, check the tunnel configuration, VPN keys, and encryption settings as well.

Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it.

Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.
ANS:

Peer Configuration Review: WAN, VPN, and NAT

Misconfiguration found: The peer router had an incorrect IP address configured as the VPN tunnel destination.

Commands used for troubleshooting:

```
None
show ip interface brief
show ip route
show interfaces tunnel 0
show crypto isakmp sa
show crypto ipsec sa
show running-config
```

```
debug crypto isakmp
```

How I identified the issue: The `show interfaces tunnel 0` command helped check the tunnel status, while `show running-config` revealed that the tunnel destination IP did not match the peer router's WAN IP. The IKE debug output helped confirm that VPN negotiation was failing.

Configuration fix:

```
None
configure terminal
interface tunnel 0
tunnel destination 198.51.100.2
end
```

Replace `198.51.100.2` with the actual WAN IP of the peer router.

Verification: I used `show interfaces tunnel 0`, `show crypto isakmp sa`, and `show crypto ipsec sa` to verify the tunnel and VPN status. I then tested connectivity between branch PCs using `ping`.

Conclusion: Correcting the tunnel destination resolved the identified configuration issue. Debugging commands helped confirm the failure rather than relying on guesswork.

On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.
ANS:

Create 3 VLANs in Cisco Packet Tracer

Objective: Create three VLANs named Insta, Zomato, and Flipkart on one switch and connect one PC to each VLAN.

1. Network topology

Cisco 2960 Switch

VLAN 10, 20, 30

PC0

Insta

192.168.10.2

PC1

Zomato

192.168.20.2

PC2

Flipkart

192.168.30.2

Connect each PC to a separate switch port using Copper Straight-Through cables.

2. VLAN configuration

Open the switch CLI and enter:

```
None
enable
configure terminal

vlan 10
name Insta
exit

vlan 20
name Zomato
exit

vlan 30
name Flipkart
exit
```

```
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit

interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
exit

interface fastEthernet 0/3
switchport mode access
switchport access vlan 30
exit

end
show vlan brief
```

3. Configure PC IP addresses

PC	Switch port	VLAN	IP address	Subnet mask
PC0	Fa0/1	Insta (10)	192.168.10.2	255.255.255.0
PC1	Fa0/2	Zomato (20)	192.168.20.2	255.255.255.0
PC2	Fa0/3	Flipkart (30)	192.168.30.2	255.255.255.0

In Packet Tracer, open each PC → Desktop → IP Configuration and enter its IP address and subnet mask. A default gateway is not required for this VLAN-only test.

4. Verify the configuration

Run on the switch:

```
None
show vlan brief
```

Confirm that Fa0/1 is in VLAN 10, Fa0/2 is in VLAN 20, and Fa0/3 is in VLAN 30.

Test connectivity from PC0:

```
None
ping 192.168.10.2
ping 192.168.20.2
ping 192.168.30.2
```

- Each PC can ping its own IP address.
- PCs in different VLANs cannot communicate by default without a router or Layer 3 switch configured for inter-VLAN routing.

Result: Three VLANs—Insta, Zomato, and Flipkart—are created on a single switch, with one PC assigned to each VLAN.

Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.
ANS:

Router-on-a-Stick Configuration — Cisco Packet Tracer

Objective: Configure one router interface with three sub-interfaces to enable communication between the Insta, Zomato, and Flipkart VLANs.

1. Network topology

Router R1

G0/0.10 — 192.168.10.1

G0/0.20 — 192.168.20.1

G0/0.30 — 192.168.30.1

802.1Q trunk link

Switch S1

VLAN 10 · Insta

VLAN 20 · Zomato

VLAN 30 · Flipkart

PC0

192.168.10.2

PC1

192.168.20.2

PC2

192.168.30.2

2. Configure router sub-interfaces

On router R1, open CLI and enter:

```
None
enable
configure terminal

interface gigabitEthernet 0/0
no shutdown
exit

interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

```
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

interface gigabitEthernet 0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit

end
write memory
```

3. Configure the switch trunk

Assume router R1 connects to switch port `Fa0/24`.

```
None
enable
configure terminal

vlan 10
name Insta
vlan 20
name Zomato
vlan 30
name Flipkart

interface gigabitEthernet 0/1
switchport mode trunk
exit

end
```

Important: The switch port connected to the router must be configured as a trunk. If your router is connected to `Fa0/24` instead, configure `interface fastEthernet 0/24` rather than `GigabitEthernet 0/1`.

4. Configure PC IP addresses and gateways

PC	IP address	Subnet mask	Default gateway
PC0 — Insta	192.168.10.2	255.255.255.0	192.168.10.1
PC1 — Zomato	192.168.20.2	255.255.255.0	192.168.20.1
PC2 — Flipkart	192.168.30.2	255.255.255.0	192.168.30.1

Set these under PC → Desktop → IP Configuration.

5. Verify the setup

On the router:

```
None
show ip interface brief
show ip route
```

On the switch:

```
None
show vlan brief
show interfaces trunk
```

From PC0, test communication with PCs in the other VLANs:

```
None
ping 192.168.20.2
ping 192.168.30.2
```

Expected result: All three VLANs can communicate through router R1, provided the trunk, VLAN assignments, IP addresses, and gateways are configured correctly.

Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command.

Hint: Use the switchport mode trunk command on the relevant interface.
ANS:

Configure and Verify a Trunk Port

1. Configure the switch port

Assume the router is connected to switch port `GigabitEthernet0/1`. On the switch CLI, enter:

```
None
enable
configure terminal
interface gigabitEthernet 0/1
switchport mode trunk
no shutdown
end
```

2. Verify trunking

Run:

```
None
show interfaces trunk
```

Expected output (example):

```
None
Port        Mode       Encapsulation  Status     Native vlan
Gi0/1       on         802.1q         trunking   1

Port        Vlans allowed on trunk
Gi0/1       1-4094

Port        Vlans allowed and active in management domain
Gi0/1       1,10,20,30
```

The exact output depends on your switch model and VLAN configuration.

3. Check the VLANs

```
None
show vlan brief
```

Confirm VLANs 10 (Insta), 20 (Zomato), and 30 (Flipkart) exist and the PC ports are assigned correctly.

Result: The router-facing switch port should show `trunking` in the `show interfaces trunk` output, allowing traffic for multiple VLANs to travel over the same physical link.

Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.
ANS:

Test Inter-VLAN Connectivity

1. Ping from Insta VLAN to Zomato VLAN

Assume:

- PC0 (Insta VLAN 10): `192.168.10.2`
- PC1 (Zomato VLAN 20): `192.168.20.2`
- Insta gateway: `192.168.10.1`
- Zomato gateway: `192.168.20.1`

On PC0, open Desktop → Command Prompt and enter:

```
None
ping 192.168.20.2
```

2. Expected result

If the configuration is correct: The ping succeeds because Router-on-a-Stick routes traffic between VLAN 10 and VLAN 20 using its sub-interfaces.

If the ping fails: Check the following:

1. Both VLANs exist and the PC switch ports are assigned to the correct VLANs.
2. The switch port connected to the router is configured as a trunk.
3. Router sub-interfaces have the correct `encapsulation dot1Q` VLAN IDs and IP addresses.
4. Each PC has the correct IP address, subnet mask, and default gateway.
5. Router interface `GigabitEthernet0/0` is enabled using `no shutdown`.

3. Verify on the router

```
None
show ip interface brief
show ip route
```

The routing table should contain connected routes for `192.168.10.0/24` and `192.168.20.0/24`.

Conclusion: The ping succeeds when the trunk, VLAN assignments, router sub-interfaces, and PC gateways are correctly configured. Without inter-VLAN routing, PCs in different VLANs cannot communicate directly.

Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.
ANS:

Add a New VLAN: Spotify (VLAN 40)

We will add Spotify as VLAN 40 while keeping the existing Insta (10), Zomato (20), and Flipkart (30) VLANs working.

1. Switch configuration

Assume `GigabitEthernet0/1` is the trunk port connected to the router and `FastEthernet0/4` connects to the new Spotify PC.

```
None
enable
configure terminal

vlan 40
 name Spotify
exit

interface fastEthernet 0/4
 switchport mode access
 switchport access vlan 40
 no shutdown
exit

interface gigabitEthernet 0/1
 switchport mode trunk
exit

end
write memory
```

2. Router configuration

Add a new sub-interface to the existing router interface `GigabitEthernet0/0`.

```
None
enable
configure terminal

interface gigabitEthernet 0/0.40
```

```
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
exit

end
write memory
```

3. Configure the Spotify PC

Open PC → Desktop → IP Configuration and enter:

Setting	Value
IP address	`192.168.40.2`
Subnet mask	`255.255.255.0`
Default gateway	`192.168.40.1`

4. Verify the configuration

On the switch:

```
None
show vlan brief
show interfaces trunk
```

On the router:

```
None
show ip interface brief
show ip route
```

Test from the Spotify PC:

```
None
ping 192.168.40.1
ping 192.168.10.2
```

The first ping checks the Spotify gateway. The second should succeed if inter-VLAN routing is configured correctly and the Insta PC is reachable.

Conclusion: VLAN 40 and sub-interface `G0/0.40` add Spotify to the existing Router-on-a-Stick setup. Existing VLANs remain unchanged, minimizing disruption. Ensure VLAN 40 is allowed on the trunk if the trunk uses a restricted allowed-VLAN list.

On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.

ANS:

Create Three VLANs on a Cisco Layer 3 Switch

Objective: Create VLAN 10 (Sales), VLAN 20 (Support), and VLAN 30 (Marketing), and assign one switch port to each VLAN.

1. Configure the switch

Open the Layer 3 switch CLI in Cisco Packet Tracer and enter:

```
None
enable
configure terminal

vlan 10
 name Sales
exit
```

```
vlan 20
 name Support
exit

vlan 30
 name Marketing
exit

interface fastEthernet 0/1
 switchport mode access
 switchport access vlan 10
exit

interface fastEthernet 0/2
 switchport mode access
 switchport access vlan 20
exit

interface fastEthernet 0/3
 switchport mode access
 switchport access vlan 30
exit

end
write memory
```

2. Connect the PCs

PC	Switch port	VLAN	IP address
PC0	Fa0/1	10 — Sales	192.168.10.2

PC1	Fa0/2	20 — Support	192.168.20.2
PC2	Fa0/3	30 — Marketing	192.168.30.2

Use subnet mask `255.255.255.0` for each PC.

3. Verify VLAN configuration

Run:

```
None
show vlan brief
```

Expected result:

- VLAN 10 — Sales — Fa0/1
- VLAN 20 — Support — Fa0/2
- VLAN 30 — Marketing — Fa0/3

Result: Three VLANs have been created successfully, with one interface assigned to each. Devices in different VLANs require inter-VLAN routing to communicate; on a Layer 3 switch, this can be configured using switched virtual interfaces (SVIs).

Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state.

Hint: Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.
ANS:

Configure SVIs on a Cisco Layer 3 Switch

Objective: Assign an IP address to each VLAN's SVI and verify that all SVIs are in the `up/up` state.

1. Configure the SVIs

Enter these commands in the Layer 3 switch CLI:

```
None
enable
configure terminal

ip routing

interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit

interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit

interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
exit

end
write memory
```

2. Assign PC IP addresses

VLAN	Department	PC IP address	Default gateway
10	Sales	192.168.10.2	192.168.10.1
20	Support	192.168.20.2	192.168.20.1

30	Marketing	192.168.30.2	192.168.30.1

Use subnet mask `255.255.255.0` for all PCs.

3. Verify SVI status

Run:

```
None
show ip interface brief
show vlan brief
show ip route
```

Expected SVI status:

Interface	IP address	Expected status
VLAN 10	192.168.10.1	up/up
VLAN 20	192.168.20.1	up/up
VLAN 30	192.168.30.1	up/up

Important: `no shutdown` enables the SVI, but does not guarantee `up/up`. Each VLAN must exist and have at least one active switch port assigned to it, with the port in a forwarding state.

4. Test inter-VLAN routing

From PC0, open Desktop → Command Prompt and run:

```
None
ping 192.168.20.2
ping 192.168.30.2
```

Expected result: Both pings succeed if the SVIs, PC IP addresses, gateways, and VLAN port assignments are correct.

Conclusion: The Layer 3 switch now has an SVI gateway for each department and can route traffic between Sales, Support, and Marketing when `ip routing` is enabled.

Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router.

Hint: Use the 'ip routing' global configuration command.

Enable Inter-VLAN Routing on a Layer 3 Switch

1. Enter the configuration commands

Open the Cisco Layer 3 switch CLI in Packet Tracer and enter:

```
None
enable
configure terminal
ip routing
end
write memory
```

The `ip routing` command enables the switch to route packets between different VLANs using their SVIs.

2. Verify the configuration

Run:

```
None
show running-config
show ip route
show ip interface brief
```

Check that the routing table contains connected routes for:

- VLAN 10 — `192.168.10.0/24`
- VLAN 20 — `192.168.20.0/24`
- VLAN 30 — `192.168.30.0/24`

3. Test connectivity

From a PC in the Sales VLAN, open Desktop → Command Prompt and run:

```
None
ping 192.168.20.2
ping 192.168.30.2
```

Expected result: Both pings succeed if the SVIs are up, the PCs have correct IP addresses and default gateways, and the switch ports are assigned to the correct VLANs.

Conclusion: Enabling `ip routing` allows the Layer 3 switch to route traffic between VLANs 10, 20, and 30 without requiring an external router.

Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

Verify Inter-VLAN Routing in Cisco Packet Tracer

1. Connect and configure the PCs

Device	VLAN	IP address	Subnet mask	Default gateway

PC0 (Sales)	10	192.168.10.2	255.255.255.0	192.168.10.1
PC1 (Support)	20	192.168.20.2	255.255.255.0	192.168.20.1

Ensure the Layer 3 switch has SVIs configured for VLANs 10 and 20, and `ip routing` is enabled.

2. Ping from PC0 to PC1

On PC0, open Desktop → Command Prompt and enter:

```
None
ping 192.168.20.2
```

Expected successful output:

```
None
Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

This is an example of expected output. Run the ping in your simulation and capture your actual result for submission. The first ping may time out while ARP resolves.

3. Verify in the switch

```
None
show ip interface brief
```

```
show ip route
show vlan brief
```

Confirm that VLAN 10 and VLAN 20 SVIs are `up/up`, both VLANs exist, and their connected routes appear in the routing table.

4. Capture the result

1. Run the ping command on PC0.
2. Ensure the successful replies and ping statistics are visible.
3. Take a screenshot of the Packet Tracer window showing the command prompt and results.
4. Submit the screenshot with your lab report.

Conclusion: Successful ping replies with `0% packet loss` demonstrate that the Layer 3 switch is routing traffic between VLAN 10 (Sales) and VLAN 20 (Support).

Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.
ANS:’

Refactor VLAN 30 to Use a /25 Subnet

Objective: Change VLAN 30 (Marketing) from `/24` to `/25` while keeping VLANs 10 and 20 unchanged.

1. Update the SVI configuration

On the Layer 3 switch, change the VLAN 30 SVI subnet mask:

```
None
enable
configure terminal
```

```
interface vlan 30
 ip address 192.168.30.1 255.255.255.128
 no shutdown
exit

end
write memory
```

The new mask `255.255.255.128` is `/25`.

2. Update VLAN 30 PC settings

Setting	Before (/24)	After (/25)
Subnet mask	255.255.255.0	255.255.255.128
Network address	192.168.30.0	192.168.30.0
Usable host range	192.168.30.1–192.168.30.254	192.168.30.1–192.168.30.126
Broadcast address	192.168.30.255	192.168.30.127
Default gateway	192.168.30.1	192.168.30.1

Update every PC in VLAN 30 to use subnet mask `255.255.255.128`. The gateway address remains unchanged.

3. Verify the configuration

Run on the switch:

```
None
show ip interface brief
show ip route
show running-config
```

Test connectivity from a VLAN 30 PC:

```
None
ping 192.168.30.1
ping 192.168.10.2
ping 192.168.20.2
```

4. Effect on communication

- VLAN 30: The subnet now supports 126 usable host addresses instead of 254.
- VLANs 10 and 20: Their `/24` configurations remain unchanged.
- Inter-VLAN communication: Communication can still work because the Layer 3 switch routes between the VLANs using their SVIs.
- Important: Devices in VLAN 30 must use the new `/25` mask. Devices using the old `/24` mask may incorrectly treat addresses such as `192.168.30.200` as local, causing connectivity problems.

Conclusion: Changing VLAN 30 to `/25` reduces its available address space without inherently disrupting communication with VLANs 10 and 20, provided the SVI and all VLAN 30 hosts use the correct subnet mask.